

A Scaling Counterexample to the Functional Gardner–Zvavitch Conjecture

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Status. Claimed full counterexample, likely valid, pending expert review. The conjecture fails already for $n = p = 1$, $t = 1/2$, standard Gaussian measure, and smooth positive even strictly log-concave functions.

1 Source question

Roysdon and Xing formulate Conjecture 6.2 in Section 6, page 29 of the arXiv PDF [1]. It asks whether an absolute constant $C \geq 1$ makes the following implication valid. For $p \in [1, \infty)$, centered integrable functions $f, g : \mathbb{R}^n \rightarrow \mathbb{R}_+$, a measure μ , $t \in (0, 1)$, and measurable $h : \mathbb{R}^n \rightarrow \mathbb{R}_+$, assume for every $x, y \in \mathbb{R}^n$ and $0 < \lambda < 1$ that

$$\begin{aligned} h((1-t)^{1/p}(1-\lambda)^{(p-1)/p}x + t^{1/p}\lambda^{(p-1)/p}y) \\ \geq f(x)^{(1-t)^{1/p}(1-\lambda)^{(p-1)/p}} g(y)^{t^{1/p}\lambda^{(p-1)/p}}. \end{aligned}$$

The proposed conclusion is

$$\int h \, d\mu \geq \frac{1}{C^p} \left[(1-t) \left(\int f \, d\mu \right)^{p/n} + t \left(\int g \, d\mu \right)^{p/n} \right]^{n/p}. \quad (1)$$

2 Proof intuition

At $p = 1$ and $t = 1/2$, the premise is a geometric-mean inequality. Consequently it is unchanged by the reciprocal amplitude rescaling

$$(f, g) \mapsto (Mf, M^{-1}g).$$

The proposed conclusion instead contains an arithmetic mean of the two integrals, which grows like $M + M^{-1}$. Starting from one centered log-concave function therefore produces an unbounded family of violations. A Gaussian profile makes every hypothesis and every integral explicit.

3 Counterexample

Theorem 1. *There is no absolute constant C satisfying Conjecture 6.2 as stated. This remains false if μ is the standard Gaussian measure on $\mathbb{R}$ and f, g, h are required to be smooth, strictly positive, even, strictly log-concave, Lebesgue-integrable, and maximized at the origin.*

bodies containing the origin, the Gardner-Zvavitch conjecture holds true up to some universal constant. This can be seen by the following Corollary 6.8.

Motivated by the advances on the Gardner-Zvavitch conjecture, we ask the following functional counterpart of this conjecture, and answer it in some special cases.

Conjecture 6.2 (Functional L_p -Gardner-Zvavitch conjecture). *Let $p \in [1, \infty)$. Let $f, g: \mathbb{R}^n \rightarrow \mathbb{R}_+$ be centered integrable functions, and μ be measure on $\mathbb{R}^n$, and $t \in (0, 1)$. Assume that $h: \mathbb{R}^n \rightarrow \mathbb{R}_+$ is a measurable function such that*

$$h((1-t)^{\frac{1}{p}}(1-\lambda)^{\frac{p-1}{p}}x + t^{\frac{1}{p}}\lambda^{\frac{p-1}{p}}y) \geq f(x)^{(1-t)^{\frac{1}{p}}(1-\lambda)^{\frac{p-1}{p}}} g(y)^{t^{\frac{1}{p}}\lambda^{\frac{p-1}{p}}}$$

holds for all $x, y \in \mathbb{R}^n$ and for all $0 < \lambda < 1$. Does there exist an absolute constant $C \geq 1$ such that following inequality hold:

$$\int_{\mathbb{R}^n} h(x) d\mu(x) \geq \frac{1}{C^p} \left[(1-t) \left(\int_{\mathbb{R}^n} f(x) d\mu(x) \right)^{p/n} + t \left(\int_{\mathbb{R}^n} g(x) d\mu(x) \right)^{p/n} \right]^{n/p} ?$$

Figure 1: Conjecture 6.2 as printed in the source paper [1, p. 29].

Proof. Fix $n = p = 1$ and $t = 1/2$, and let $\mu = \gamma_1$ be standard Gaussian measure. For an amplitude $M > 0$, define

$$f_M(x) = Me^{-x^2}, \quad g_M(x) = M^{-1}e^{-x^2}, \quad h(x) = e^{-x^2}.$$

These functions have all of the regularity and shape properties asserted in the theorem. In particular they are even, so their barycenters vanish, and each attains its maximum at the origin. Thus the example is centered under both standard readings relevant to the source statement.

For $p = 1$, the factors containing $\lambda^{(p-1)/p}$ and $(1-\lambda)^{(p-1)/p}$ equal one. At $t = 1/2$ the pointwise hypothesis becomes

$$h\left(\frac{x+y}{2}\right) \geq \sqrt{f_M(x)g_M(y)}.$$

For every $x, y \in \mathbb{R}$,

$$\begin{aligned} h\left(\frac{x+y}{2}\right) &= \exp\left(-\frac{(x+y)^2}{4}\right), \\ \sqrt{f_M(x)g_M(y)} &= \exp\left(-\frac{x^2+y^2}{2}\right). \end{aligned}$$

The desired comparison follows from the exact identity

$$\frac{x^2+y^2}{2} - \frac{(x+y)^2}{4} = \frac{(x-y)^2}{4} \geq 0.$$

It holds for every $0 < \lambda < 1$, since λ has disappeared from the $p = 1$ specialization.

Set

$$A := \int_{\mathbb{R}} e^{-x^2} d\gamma_1(x) = \frac{1}{\sqrt{3}} > 0.$$

Then

$$\int h d\gamma_1 = A, \quad \int f_M d\gamma_1 = MA, \quad \int g_M d\gamma_1 = M^{-1}A.$$

If (1) held with an absolute constant C , it would imply

$$A \geq \frac{1}{C} \left(\frac{MA}{2} + \frac{M^{-1}A}{2} \right), \quad \text{hence} \quad 1 \geq \frac{M + M^{-1}}{2C}.$$

For any fixed C , this is false when M is sufficiently large (for example, $M > 2C$). Therefore no such absolute constant exists. $\square$

4 Structural diagnosis and scope

The obstruction does not use singular measures, rough functions, lack of symmetry, or tail pathologies. It is the mismatch between two homogeneities: the premise is invariant under reciprocal rescaling at $p = 1, t = 1/2$, while the claimed conclusion is not. The source’s nearby Theorem 6.3 assumes $\|f\|_\infty = \|g\|_\infty$, and its later functional estimate normalizes each integral by the corresponding supremum. Such hypotheses eliminate this scaling family.

This packet disproves only Conjecture 6.2 exactly as printed. It does not disprove versions imposing equal heights, equal masses, probability normalization, or scale-invariant normalized integrals. It also says nothing negative about the geometric Gardner–Zvavitch conjecture for convex bodies.

5 Verification and novelty check

The accompanying verifier reduces the pointwise inequality to $(x - y)^2/4 \geq 0$, evaluates the Gaussian integral exactly as $1/\sqrt{3}$, and checks explicit violating amplitudes. These symbolic checks are confirmations rather than ingredients of the proof.

A bounded literature search on 9 August 2026 used the exact conjecture name and wording, arXiv:2004.09737, the Roysdon–Xing title and visible citation neighborhood, the corresponding 2020 dissertation statement, and later functional Gardner–Zvavitch work. It found the same conjecture statement and normalized or geometric variants, but no matching reciprocal-scaling counterexample. Novelty is therefore plausible, not certified.

The main reviewer priority is semantic: confirm that the source’s word “centered” carries no hidden amplitude normalization. The printed conjecture states none, and the example is centered both by barycenter and by location of its maximum.

References

- [1] M. Roysdon and S. Xing, *On L_p -Brunn–Minkowski type and L_p -isoperimetric type inequalities for measures*, Trans. Amer. Math. Soc. **374** (2021), 5003–5036; arXiv:2004.09737v2; doi:10.1090/tran/8356.